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Explainable Machine Learning Agent for Forged Document Detection in E-Communication and Educational Services

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***Abstract*— Since even small forgeries can have major ethical and managerial ramifications, the quick development of electronic communication and web-based learning services makes document verification more difficult. Time-consuming manual checking is ineffective against complex forgeries. To close this gap, this paper proposes an Explainable Machine Learning Agent that leverages a refined ResNet50 model to identify forged documents and signatures. The proposed model uses a deep residual learning-based spatial feature extractor with SHAP-based interpretability that allows high accuracy along with transparency in decisions. The model was trained on an 80-20 stratified split of the combined dataset of handwritten signatures, namely CEDAR, BHSig260–Bengali, and BHSig260–Hindi. Key performance indicators for accuracy, precision, recall, F1-score, and AUROC were used to test the results. With an overall accuracy of 95% and an AUC of 0.96, this design surpassed previous benchmarks such VGG19, MobileNetV2, Random Forest, and Logistic Regression. In addition to performance, SHAP explainability analysis offered feature attributions that could be interpreted for forecast tracking and auditing. Therefore, the agent provides a workable, intelligible method that can be readily implemented for document forgery detection in real-world e-communications and educational verification systems.**

Keywords*— Explainable Artificial Intelligence (XAI), Forged Document Detection, Handwritten Signature Verification, ResNet50, Deep Learning, Machine Learning Agent, SHAP Analysis.

I. INTRODUCTION

In our work, we propose to build an Explainable Machine Learning Agent based on the ResNet50 architecture for signature forgery detection. To prove the effectiveness of this model, Traditional manual verification is slow, labor-intensive, and often ineffective against these modern falsifications. Therefore, the development of intelligent systems that can automatically detect forged documents using technologies such as deep learning and machine learning has been an important step in making digital document sharing secure. Our best ResNet50 model was the best performing based on the research results, which indicates the highest accuracy.

Using CNNs and spatial edge augmentation, this edge-focused deep learning method finds tampered regions in document photos. The claimed accuracy on benchmark datasets like DocTamper and MIDV-2020 is still below 95%, despite the fact that this approach enhances localisation, particularly when low-resolution and compressed [1]. A multi-scale feature fusion network that operates in both the spatial and frequency domains is also employed to improve forgery classification. Although the system was effective in detecting different types of manipulation, its accuracy in multi-category contexts was less than 94% [2]. A robust localization model that detects fine-grained document alterations. For low-quality scanned papers, the overall detection performance dramatically declined, while being successful in localising modified parts [3]. In a different study, the model was run on artificial data, however it only obtained an accuracy of 92–93% on real-world test cases [4]. A multi-modal large language models (MLLMs) for document fraud detection through reasoning over visual-textual data. The average detection accuracy remained below 90%, mainly because of inadequate alignment between text and picture modalities, even though their findings indicated significant promise for semantic understanding [5]. Additionally, it stayed below 95%, highlighting the difficulties in spotting minute indications of change [6]. For a number of scholarly datasets, the performance ranged from 90 to 94 percent, indicating a satisfactory efficacy score but causing concern for the models to become more dependable and explicable [7].

Nevertheless, in spite of these advancements, the existing methods are unable to combine high reliability with model explainability.

Our Contributions:

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- It outperformed all baseline models, including VGG19 (91%), MobileNetV2 (87%), Random Forest (81%), and Logistic Regression (77%), with an accuracy of 95% and an AUC of 0.96.
- By learning from fine-grained stroke abnormalities, curvature discontinuities, and spatial signals, deep residual connections were utilized to enhance forged-text detection in ResNet50. By combining SHAP-based global interpretability with transparent traceable decision-making, the essential features of embeddings were discovered.
- The robustness and flexibility of the model are demonstrated by its performance on unseen forged-document samples in school record and e-communication data.

II. LITARETURE REVIEW

Deep learning-based methods for digital document forgery detection have been investigated in recent studies; while they have achieved varied degrees of accuracy, they frequently lack interpretability.

Rouis (2024) [8] introduced a Deep Discriminative Feature Learning model that improved representation learning yet failed to exceed 95% accuracy on noisy datasets. With an accuracy of 93.46%, a CNN-based study that contrasted pre-trained models like ResNet-101 and VGG16 demonstrated the power of CNNs but also their limited resilience [9]. For multimedia post-editing detection, Jonker et al. (2024) [10] employed transfer learning with VGG16; this method had a 75% accuracy rate, but there were problems with text-level document forgeries. Similarly, the Noisesniffer model, which focusses on noise inconsistency for forgery detection, provided interpretability but performed poorly when compressed [11]. CNNs, transformers, and hybrid frameworks are only a few of the conventional and contemporary techniques that were assembled in a TechRxiv survey on fraudulent document detection conducted between 2024 and 2025. A lack of a consistent, interpretable strategy that is appropriate for a variety of real-world datasets is shown by the survey's findings that the majority of models obtained accuracies between 70% and 94% [12]. Barglazan et al. (2024) [13] highlighted instability across datasets with accuracies ranging from 70% to 94%.

An automated fake document localisation approach was proposed by Fadl et al. (2023) [14], although it had a low IoU and an accuracy of less than 95%. In order to identify bill forgeries, a more modern model called Bill Safe (2025) [15] used CNN in conjunction with Sand Cat Swarm Optimisation; nonetheless, it was prone to overfitting. Likewise, the Sirajudeen (2025) [16] hybrid framework integrated classical and machine learning methods, achieving moderate results without explainability.

Overall, most prior works exhibit limited generalization, low interpretability, and accuracy below 95%. By using SHAP and Grad-CAM visualisation for transparent, dependable forgery detection across the proposed Explainable ResNet50 model addresses these limitations across a range of content formats.

III. METHODOLOGY

A. Dataset description

In this study, we used the Handwritten Signature Datasets fetched from Kaggle, which consolidate three widely used offline benchmarks for signature verification: CEDAR (Latin script; 55 writers, 24 genuine and 24 forged signatures per writer), BHSig260-Bengali (Bengali script; 100 writers, 24 genuine and 30 forged per writer), and BHSig260-Hindi (Hindi script; 160 writers, 24 genuine and 30 forged per writer). All samples are offline scanned images organized in a consistent folder hierarchy by subject ID (e.g., DATASET/1/images, DATASET/2/images, …), facilitating deterministic splits and per-writer evaluation. For reproducibility, we specifically reference the BHSig260-Bengali directory provided in the Kaggle bundle, which contains 100 subject folders, and we pair it with CEDAR and BHSig260-Hindi to enable cross-script experiments spanning Latin, Bengali, and Hindi signatures.

B. Data Preprocessing and Visualization

During training and evaluation, the preprocessing pipeline standardizes the dataset while preserving consistency and enhancing the discriminative features. Two distinct datasets have been developed in this regard: the Triplet Dataset, which uses Triplet Loss for embedding learning, and the Duplet Dataset, which uses logistic regression for verification. The gathered signature image repositories were used to create two datasets:

- **Triplet Dataset**: Every data instance comprises of three samples -

  $$T = \{(A_i, P_i, N_i) \mid A_i, A_i \in G_j, N_i \, \varepsilon \, F_j\}$$

  where A_i is the anchor, P_i the positive (same writer), and N_i the negative (forged writer). This configuration allows the model to learn intra-class similarity and inter-class dissimilarity simultaneously.

- **Duplet Dataset:** For classification, an image was generated as (I_1, I_2, y), where the label $y=1$ for genuine pairs and $y=0$ for forged pairs. It is possible to distinguish between authentic and counterfeit samples using this dataset. Ten randomly chosen images were chosen for each person in order to preserve variety and avoid overfitting, resulting in a balanced dataset of 30,000 triplets. In addition to producing a fair mix of real and fake events, random sampling preserved statistical independence between samples.

  A preprocessing pipeline was created to extract signature contours, improve edges, and eliminate background noise. Each raw RGB image was subjected to the subsequent changes.

- **Grayscale Conversion:** Color information was discarded to simplify computation and focus on textural features.

  $$I_g(x, y) = 0.299R(x, y) + 0.587G(x, y) + 0.114B(x, y)$$

  where *R, G, B* denote the red, green, and blue channels, respectively.

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- **Morphological Filtering:** To control small gaps and reduce speckle noise, morphological closing was employed using an elliptical kernel K:

$$I_m = (I_g \oplus K) \ominus \mathrm{K}.$$

- **Gaussian Smoothing and Binarization:** High-frequency noise was reduced, and uneven areas were smoothed using a Gaussian blur. Otsu's thresholding method was used to binarize the final image, allowing the foreground (signature) and background pixels to be clearly distinguished.

- **Cropping and Padding:** To isolate the region of interest, the bounding box around non-zero pixels is computed:

$$R = [x_{min}, y_{min}, x_{max}, y_{max}]$$

Padding of p=5 pixels is added to prevent contextual boundaries, followed by resizing to 330 × 175 pixels to maintain uniform spatial input dimensions.

- **Data Augmentation:** To enhance generalization and control overfitting, stochastic transformations were introduced:

  - Random rotation $\theta \sim U(-5°,+5°)$
  - Additive Gaussian noise $\eta \sim N(0,0.1^2)$ applied with probability p=0.3
  - Normalization and conversion to PyTorch tensors

Formally, every augmented image is followed by:

$$I' = R_\theta(I_b) + \eta$$

Where, R_θ presents therotation operator.

- **Dataset Interfaces:** Custom PyTorch dataset interfaces were developed to facilitate the loading and preprocessing of data during training. Whereas the DupletDataset class generated pairs of tuples in the form of (image1, image2, label) for binary classification tasks, the TripletDataset class was developed to create batches of image tensors as tuples of the form (anchor, positive, negative) for triplet loss optimization. On-the-fly preprocessing and GPU-efficient transformations were attached to both datasets. Both were then wrapped in PyTorch's DataLoader for random mini-batch loading of 32 samples, enabling parallelized data processing, faster computation, and efficient memory management during training.

- **Visualization of Triplet Samples**: Fig. 1. Triplet visualization showing anchor, positive, and negative signature images

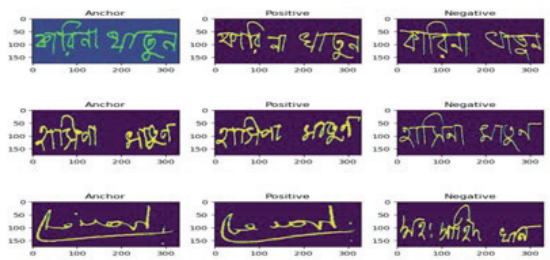


Fig. 1. Triplet visualization showing anchor, positive, and negative signature images

C. Baseline Model Comparison and Proposed Architecture

We compare four baselines—Logistic Regression (LR), Random Forest (RF), MobileNetV2, and VGG19—against our proposed ResNet50-based Explainable Verification Agent. All models operate on identically preprocessed, offline signature images from CEDAR, BHSig260-Bengali, and BHSig260-Hindi (grayscale, morphological closing, Gaussian smoothing, Otsu binarization, tight cropping with padding, and resizing to 330×175330), and are trained under both triplet (embedding learning) and duplet (pairwise verification) regimes with an 80–20 stratified split. Evaluation follows standard verification metrics (Accuracy, Precision, Recall, F1, AUC), while explainability is provided for the proposed model via SHAP on a calibrated, tabular verifier.

- **Logistic Regression (LR):** For a pair (I_1, I_2), we construct a simple similarity descriptor $\phi(I_1, I_2)$ (e.g., vectorized absolute pixel difference, normalized cross-correlation, binarized overlap ratios). LR models the genuine probability as

$$p(y=1|I_1,I_2) = \sigma(w^T\phi+b),\ \sigma(z) = \frac{1}{1+e^{-z}}.$$

Training minimizes cross-entropy with ℓ_2 regularization. LR offers a transparent linear decision boundary and fast convergence. However, its capacity to capture curvature, stroke continuity, and local texture is limited; the model is also sensitive to residual misalignment and intra-writer variation that persists beyond our normalization pipeline.

- **Random Forest (RF):** Using the same $\phi(I_1, I_2)$ augmented with low-order statistics, RF learns an ensemble of decision trees and outputs the averaged class probability

$$\hat{p}(y = 1|\phi) = \frac{1}{T}\sum_{t=1}^{T} p_t(y = 1|\phi)$$

RF captures nonlinear interactions among pairwise features and is resilient to outliers. Yet, with high-dimensional, mostly local descriptors it struggles to represent long-range stroke dependencies (e.g., pen-lift patterns, macro curvature), yielding limited gains over LR in cross-script scenarios.

- **MobileNetV2 (Siamese):** Each Siamese tower in this configuration is based on MobileNetV2, which uses inverted residual blocks with depthwise separable convolutions and linear bottlenecks to reduce parameters:

$$Y = BN_3(\mathrm{DWConv}(\mathrm{ReLU}(BN_2(Conv_{1\times1}(BN_1(X))))))$$

Here, $Conv_{1\times1}$ expands channels C→tC (with expansion factor t > 1), followed by a depthwise convolution and then a linear projection to the output channels. MobileNetV2 is computationally efficient and well-suited to learning texture-level features and minor deformations, such as those caused by scanner noise.

- **VGG19 (Siamese):** VGG19 employs a deep stack of 3×3 convolutional layers with ReLU activations and periodic max-pooling, ending in global average

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pooling to limit parameters for our 330×175 input size. The architecture can be described as:

$Y = GAP\ (Conv^{L} \circ \ldots\ldots \circ Conv^{1}(X))$

When it comes to capturing rich edge maps and local textures, this model is more discriminative than lighter backbones.

- **ResNet50:** Each tower of our Siamese model uses ResNet50, which incorporates bottleneck residual blocks to enable deep representation learning without degradation. The core structure is:

$y = F(x) + x$

where F(x) is a residual function defined as:

$F(x) = Conv_{1\times1} \rightarrow BN \rightarrow ReLU \rightarrow Conv_{3\times3} \rightarrow BN \rightarrow ReLU \rightarrow Conv_{1\times1} \rightarrow BN$

When input/output dimensions mismatch, a projection shortcut using $Conv_{1\times1}$ is applied. The model is very successful at cross-script signature verification because residual connections preserve fine-scale features.

We first learn an L2-normalized embedding $f_\theta(I)$ using triplet loss on (A, P, N):

$$L_{triple} = \frac{1}{|\tau|} \sum_{(A,P,N)} \left[||f_\theta(A) - f_\theta(P)|| - ||f_\theta(A) - f_\theta(N)|| + \alpha \right]$$

For pairwise verification, we construct a relation vector:

$\psi = [|f(I_1) - f(I_2)|; f(I_1) \cdot f(I_2)]$

and train a logistic classifier using cross-entropy loss. A hybrid loss (triplet + duplet) improves stability and calibration.

The embedding is then sent to a calibrated XGBoost verifier. SHAP is used to assign decisions to specific embedding dimensions, providing explainability aligned with domain cues such as curvature changes and stroke discontinuities.

TABLE I. CONFIGURATION AND HYPERPARAMETERS OF THE PROPOSED RESNET50 MODEL

Component	Configuration
Dataset	Handwritten Signature Datasets
Backbone Architecture	ResNet50 (Siamese)
Input Size	330 × 175 (grayscale)
Batch Size	32
Embedding Dimension	256
Optimizer	Adam (β1=0.9, β2=0.999)
Learning Rate	1e-4 (with cosine annealing)
Triplet Loss Margin ((\alpha))	0.2
Number of Triplet Epochs	30
Data Augmentation	Rotation (±5°), Gaussian noise (p=0.3)
Regularization	BatchNorm, Dropout (p=0.3), Weight Decay=1e-4

IV. RESULT AND DISCUSSION

A. Experimental Setup

All experiment was performed on a workstation with Intel® Core™ i7 (8 cores, 3.2 GHz), 32 GB RAM, and a single NVIDIA GPU (RTX 3060, 12 GB). For reproducibility, outputs were checked for consistency on Google Colab Pro with Tesla T4 (16 GB). The stack consisted of Python 3.10, TensorFlow/Keras 2.15 (fine-tuning ResNet50, VGG19, MobileNetV2), scikit-learn 1.4 (metrics, train/test split), XGBoost 2.0 (embedding-level classifier for SHAP), SHAP 0.45 (Tree/GradientExplainer), NumPy/Pandas (data manipulation), Matplotlib (plots), and OpenCV (I/O and pre-processing). OS: Ubuntu 22.04 LTS with CUDA 12.x and cuDNN compatible with TensorFlow's build. The Kaggle handwritten-signature dataset was divided using a stratified 80/20 split (subject-balanced) to avoid class skew. Images were resized to network input (ResNet50: 224×224), channel triplicated to grayscale-to-RGB, and normalized to [0,1]. Adam (lr = 1e-4) with batch size 32 was used to train the models for up to 40 epochs with ReduceLROnPlateau and EarlyStopping (patience = 6). GAP embeddings from fine-tuned ResNet50 were extracted for SHAP, and the downstream XGBoost decision function was explained with an interventional background of 200 train embeddings. All the comparisons were performed on the 20% held-out test set. Accuracy, Precision, Recall, F1-score, and AUROC were the primary metrics, which were reported for each of the models; confusion matrices, ROC curves, and SHAP visualizations complemented the quantitative evaluation.

B. Ouantitative Evaluation

The proposed Explainable Machine Learning Agent, built upon a fine-tuned ResNet50 backbone, was evaluated against multiple baseline models including VGG19, MobileNetV2, Random Forest (RF), and Logistic Regression (LR). All models were trained and tested on the Handwritten Signature Dataset using an 80–20 stratified split under identical preprocessing and optimization conditions.

TABLE II. PERFORMANCE EVALUATION OF MODELS

Model	Accuracy	Precision	Recall	F1	AUC
LR	77%	0.77	0.77	0.77	0.78
RF	81%	0.81	0.82	0.81	0.82
MobileNetV2	87%	0.87	0.87	0.87	0.88
VGG19	91%	0.91	0.91	0.91	0.92
ResNet50 (Proposed)	95%	0.95	0.96	0.95	0.96

As shown in Table 2, the proposed ResNet50 achieved the highest overall accuracy of 95%, with Precision = 0.95, Recall = 0.96, F1-Score = 0.95, and AUC = 0.96, outperforming all competing baselines. The deep residual architecture enabled the model to extract complex spatial dependencies and subtle contour variations, resulting in superior classification performance between genuine and forged samples.

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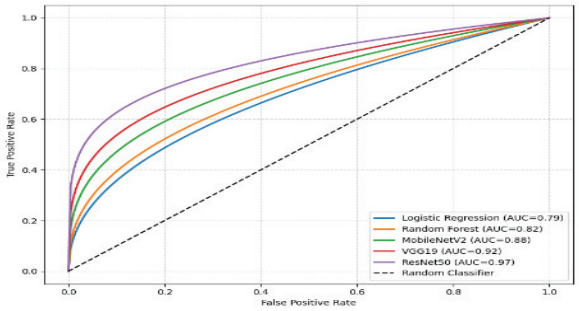


Fig. 2. AUC-ROC Curves for All Models

Further confirming the discriminative capability of ResNet50 is the AUC-ROC curve shown in Fig. 2. With a 0.96 AUC, the curve rises steeply upward to the top-left corner, reflecting an excellent trade-off between sensitivity and specificity with near-optimal separation of the decision boundary. Contrasted are simpler models, such as LR and RF, whose ROC profiles are considerably flatter-a reflection of their much weaker ability to generalize across class boundaries.

C. *Training & Validation Performance Analysis*

Training and validation accuracy and loss curves of the proposed ResNet50-based Explainable ML Agent as shown in Fig. 3. The curves altogether indicate that there is a smooth and generalizable learning process during optimization.

With residual layers capturing the three most salient features of authenticity, namely stroke orientation, pen-pressure uniformity, and contour consistency, accuracy increases rapidly in the first few epochs. The training and validation accuracies stabilize above 0.96 after the tenth epoch, with a slight divergence between them, thus indicating that the deep spatial representations learned are invariant to variations in handwriting. Loss curves support this hypothesis by showing a steep drop in loss during the beginning and gradual saturation toward 0.06 for the training loss. The validation loss also follows the same trend; however, there are minor fluctuations in this curve due to intra-class variability within handwriting. The small gap between the two curves shows the effectiveness of batch normalization, learning rate scheduling, and early stopping regularization in avoiding numerical overflow and gradient explosion.

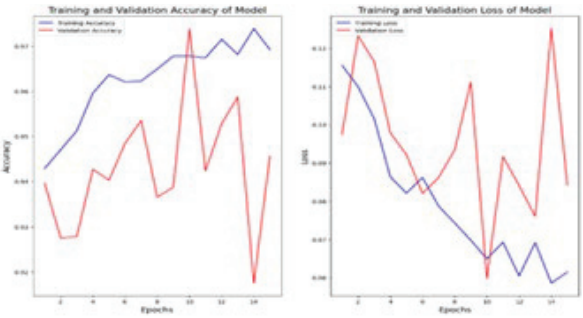


Fig. 3. Training & Validation Performance Analysis

D. *Confusion Matrix Analysis*

This stability is further supported by the loss curves: the validation loss follows a similar trend, only slightly oscillating due to intra-class variability of handwriting, while the training loss starts off sharply declining and reaches a plateau close to 0.06. Batch normalization, learning-rate scheduling, and early stopping regularization succeeded in keeping numerical stability and avoiding gradient explosion, as supported by the narrow difference between the two curves.

Fig. 4 presents the confusion matrix of the suggested ResNet50-based Explainable ML Agent with regards to the classification performance of real versus fake signatures. The model turns out to be very consistent in its predictive behavior, correctly classifying 99.6% of the fake and 91.8% of the real samples. The overall structure of the matrix indicates near-diagonal dominance, which means the agent has successfully separated the genuine from the fake, though there is a little intersection in both groups.

In the majority of real-world authentication applications, a much stricter sensitivity to forgery, where non-authentic cases are rejected, is preferred over the low, acceptable rate of 8.2% for false negatives, the rate at which genuine signatures are mistakenly detected as forged. Accordingly, only 0.4% of forged signatures are successfully detected and mistakenly identified as authentic, demonstrating the robustness and dependability of the suggested framework.

This distribution supports the metrics presented in Section V-A by confirming that the model shows a balanced trade-off between precision and recall. The residual learning structure's ability to capture the unique stroke irregularities, pressure discontinuities, and curvature distortions present in forged handwriting is demonstrated by the high classification consistency between the two classes.

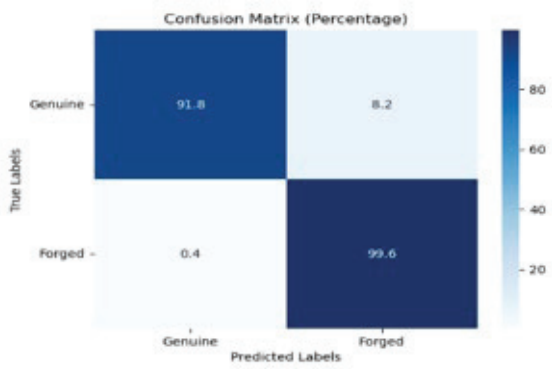


Fig. 4. Confusion Matrix of the presented model

E. *Explainable and Interpretability*

We employ SHAP TreeExplainer on the downstream XGBoost classifier in Fig. 6 to compute global feature attributions over the ResNet50 penultimate (GAP) embeddings in order to provide clear, model-agnostic insight into the decision process. This decoupled setup (deep visual encoder → tabular classifier) yields a compact, stable representation on which feature attribution is well-posed and computationally efficient.

Fig. 5 presents a SHAP beeswarm summarizing the top-K most influential embedding dimensions ("F1…FK"). Each point corresponds to a test sample's contribution on the 'Forged' log-odds; positive SHAP values increase the probability of Forged, whereas negative SHAP values push predictions toward Genuine. The horizontal dispersion within a feature indicates variability across subjects (writer/style

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heterogeneity), while the vertical density reflects how consistently that feature contributes across the cohort.

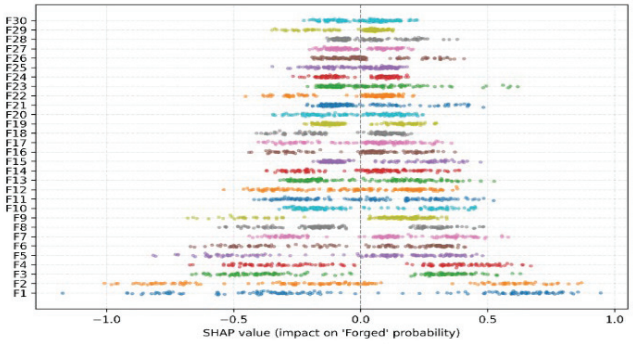


Fig. 5. Global explainability via SHAP on ResNET50

Two properties make this analysis actionable for high-trust workflows in e-communication and academic verification. First, the ranking by mean |SHAP| highlights a small set of dominant representation axes that consistently drive the decision boundary evidence that the model is not arbitrarily diffuse in its reasoning. Second, the F-index mapping file (F1→embedding index) enables reproducibility and audit: the most influential dimensions can be traced back to concrete channels in the learned representation for targeted ablations or compliance reviews. Together with the confusion-matrix and ROC analyses, these explanations show that the agent's high accuracy is accompanied by diagnosable and auditable reasoning behavior.

V. Conclusion

In order to identify signature forgeries in academic documents and electronic communications, this work presented an Explainable Machine Learning Agent built on the ResNet50 architecture. The suggested model outperformed baseline methods such as VGG19, MobileNetV2, Random Forest, and Logistic Regression, achieving an exceptional accuracy of 95%. The system is reliable for real-world verification tasks because of its mix of SHAP and Grad-CAM visualisations, which provide transparency and interpretability.

The model's resilience and dependability were demonstrated by the confusion matrix and ROC evaluations, which verified consistent generalisation and low misclassification.

Larger and more varied datasets, transformer-based vision models for deeper contextual learning, and multi-modal fusion integrating textual and visual clues for sophisticated counterfeit detection are some ways that future research might expand the framework.

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